



Matteo Merler

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Summary

I am a pre-doctoral researcher at FBK NLP in Trento, Italy, with an MSc in Machine Learning, Data Science, and AI from Aalto University in Helsinki. My research focuses on **bridging large-scale models** (such as LLMs and VLMs) **with methods like Reinforcement Learning (RL) and Symbolic Planning**. I aim to explore how RL's ability to learn from experience can complement the strengths of LLMs, which currently lack this capacity. I'm particularly interested in **model-based RL and world models**, investigating how large-scale models can be trained to plan ahead and imagine future states, helping them to adapt to complex reasoning and long-horizon planning beyond pattern matching.

Education

- M.Sc. [Aalto University](#)** , Machine Learning, Data Science and Artificial Intelligence Sept 2022 – Sept 2024
- Graduated with honors, with a **final grade of 5.0/5.0** and a **GPA of 5.0/5.0**.
 - Thesis: **[Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search](#)** , supervised by Prof. Pekka Marttinen, published at NeurIPS 2024.
 - Thesis awarded as one of the three **best MSc theses of 2024** at the School of Science.
 - Received the Dean's Scholarship for outstanding academic performance.
- B.Sc. [University of Trento](#)** , Computer Science Sept 2019 – July 2022
- Graduated with a **final grade of 110/110 cum laude (with honors)** and a **GPA of 28.46/30**.
 - Thesis: **[Route Optimization for Indoor Spaces](#)**, supervised by Prof. Alberto Montresor (in Italian).
- H.S. [Liceo Scientifico Leonardo da Vinci](#)** , Liceo Scientifico (Italian Scientific High School Diploma) Sept 2014 – July 2019
- Graduated with a **final grade of 100/100**.

Experience

- Researcher, [Fondazione Bruno Kessler](#)** – Trento, Italy Jan 2025 – present
- Working as a pre-doctoral researcher ("Researcher 4 contract") in the [Natural Language Processing Lab](#) under the supervision of Bernardo Magnini.
- Lead researcher on FBK's side of a joint project on **planning with Vision-Language Models** (with Aalto University), under resubmission.
 - Led a second project on **selectively integrating VLM guidance with model-free Reinforcement Learning** based on the agent's own uncertainty, with a preliminary version accepted at the CAIPI workshop at ECAI 2025 and a full paper in preparation.
 - Participating on private research for collaborations with companies.
 - Contributing to the draft of a **Horizon Europe funding proposal**, leading the lab's team for Language and Reinforcement Learning.
 - Active participation to the research group's scientific activities, including monthly meetings and collaborative discussions.
- Research Assistant, [Aalto University](#)** – Helsinki, Finland Aug 2024 – Sept 2024
- MSc Thesis worker in the [Foundation Models for Language and Reinforcement Learning](#) (Larel) group, supervised by Prof. Pekka Marttinen.
- Conducted a research project on **Symbolic Regression with Large Language Models**, presented at the ACL Student Research Workshop.

- Co-led a research project on **Code World Models** with PhD candidate Nicola Dainese, forming the basis of my Master's Thesis, accepted at NeurIPS 2024.
- Started a project on **planning with Vision-Language Models**, later continued by both Aalto University and my current group at FBK.
- Regularly attended lab meetings and participated in broader academic activities, including the Seminar on Large Language Models.

Internship, [ThinkIN](#)  – Trento, Italy

Feb 2022 – June 2022

Worked as a software developer intern in collaboration with the University of Trento.

- Designed and built a prototype for an indoor navigation system specifically targeted for supermarkets, to be used with the company's own location tracking system.

Publications

Guiding Reinforcement Learning with Selective Vision-Language Model Supervision 

Oct 2025

Matteo Merler, Giovanni Bonetta, Bernardo Magnini

ECAI Workshop on AI-based Planning for Complex Real-World Applications (CAIPI'25)

ViPlan: A Benchmark for Visual Planning with Symbolic Predicates and Vision-Language Models 

May 2025

Matteo Merler, Nicola Dainese, Minttu Alakuijala, Giovanni Bonetta, Pietro Ferrazzi, Yu Tian, Bernardo Magnini, Pekka Marttinen
arXiv preprint

Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search 

Dec 2024

Nicola Dainese, **Matteo Merler**, Minttu Alakuijala, Pekka Marttinen

Advances in Neural Information Processing Systems 37 (NeurIPS 2024)

In-Context Symbolic Regression: Leveraging Large Language Models for Function Discovery 

Aug 2024

Matteo Merler, Katsiaryna Haitsiukevich, Nicola Dainese, Pekka Marttinen

Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, Volume 4: Student Research Workshop (ACL 2024 SRW)

Additional Experience and Awards

Master's Thesis Award : My Master's thesis was awarded as one of the three best MSc theses of 2024 at the School of Science.

Volunteer

Volunteer, [Libera Associazioni Nomi e Numeri Contro le Mafie](#)  – Calabria, Italy

July 2017 – July 2017

I participated in a volunteering camp experience in southern Italy where we cultivated the fields of mafia-confiscated houses for the Libera association, while also taking part in various activities on the territory and listening to testimonies from victims and policemen.

Volunteer, [Associazione Volontaria "Per un Mondo Migliore"](#)  – Rijeka, Croatia

Aug 2018 – Aug 2018

I participated in a volunteering camp experience in Rijeka, Croatia where we helped different families and associations in need around the city and in mainland Croatia while living together in a shared home.

Languages

Italian: Native

English: C2 (Near-native proficiency)

German: A2 (Basic)